

Salesforce Documentation

Business Rules Engine

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Generated: 2026-01-27

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notes.rn_business_rules_engine_intro.htm&release=244&type=5](https://help.salesforce.com/s/articleView?id=release-notes.rn_business_rules_engine_intro.htm&release=244&type=5)

Business Rules Engine

To understand the reasoning behind the output of an expression set that has Decision Explainer enabled, capture action logs each time the expression set is run from Salesforce Flow. To simulate an expression set version that has a large number of input variables, pass the variable values in JSON format. Export decision matrix versions to DMN files and reuse them outside Business Rules Engine.

- **Export Decision Matrix Versions for Reuse Outside Business Rules Engine**

Make it easier to collaborate on business decisions by exporting decision matrix versions to Decision Model and Notation (DMN) files. DMN is a standardized specification for writing business rules. With DMN files, you can reuse a matrix version created in Business Rules Engine in any tool that supports DMN files. This release supports exporting matrix versions to DMN files only through connect APIs. The exported files can be downloaded from a decision matrix version's record detail page in the Business Rules Engine app.

- **Use Currency and Date/Time Data Types in Decision Tables**

Create a wider range of rules in decision tables using the Currency and Date/Time data types. With the support for the Currency data type, you can now look up decision tables that use the Currency data type from expression set versions.

- **Clone Expression Sets**

If you want to modify an expression set's versions without affecting the original expression set, clone the expression set. Cloning an expression set clones all the versions in the expression set. You can copy the versions of an expression set that's used as a subexpression in another expression set or is referenced in a flow, without breaking the calling expression set or flow.

- **Capture Action Logs for Expression Sets Called from Salesforce Flow**

When a flow calls an expression set that has Decision Explainer enabled, action logs are now generated for every run of the expression set. Action logs show the rationale behind the expression set's output. To capture action logs, you specify the record ID of the object for the `__actionContextCode` parameter, and the explainability action definition or the purpose for which the expression set is used for the `__specName` parameter.

- **View Business Elements as First-Level Elements in Expression Set Builder**

Business elements shipped with your cloud are now listed as first-level elements and look like standard step elements. Previously, you had to add a business element as a step on the builder canvas, and then search for and select the business element from the available list.

- **Simulate Expression Set Versions Using JSON Input**

Want to simulate an expression set version that has a large number of input variables? You can now pass the input data in JSON format using the Advanced input mode for simulation. With the Advanced input mode, you can enter the input values in the JSON input text area. Previously, all the input variables in the version only appeared as fields, and you entered values for each field. The JSON input option simplifies and speeds up the

process of passing input variables, especially when you have a large number of values to input.

- **Changed Object in Business Rules Engine**

Do more with this changed Business Rules Engine object.

- **Connect REST APIs**

Connect REST APIs for Business Rules Engine help customers, partners, and ISVs integrate with Salesforce software and UIs.

- **New and Changed Business Rules Engine Metadata Types**

Make the most of the new and changed metadata types of Business Rules Engine.

Source: [https://help.salesforce.com/s/articleView?id=release-
notes.rn_export_matrix_version_dmn_file.htm&language=en_US&release=244&type=5](https://help.salesforce.com/s/articleView?id=release-notes.rn_export_matrix_version_dmn_file.htm&language=en_US&release=244&type=5)

Export Decision Matrix Versions for Reuse Outside Business Rules Engine

Make it easier to collaborate on business decisions by exporting decision matrix versions to Decision Model and Notation (DMN) files. DMN is a standardized specification for writing business rules. With DMN files, you can reuse a matrix version created in Business Rules Engine in any tool that supports DMN files. This release supports exporting matrix versions to DMN files only through connect APIs. The exported files can be downloaded from a decision matrix version's record detail page in the Business Rules Engine app.

Where: This change applies to Lightning Experience in Enterprise, Unlimited, and Developer editions for Industries clouds where Business Rules Engine is enabled.

See Also

[Salesforce Help: Download Matrix Versions Exported to DMN Files](#)

[Salesforce Help: Limitations for Exporting Matrix Versions to DMN Files](#)

Source: https://help.salesforce.com/s/articleView?id=sf.download_matrix_versions_exported_to_dmn_files.htm&language=en_US&type=5

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Export and Download Matrix Versions

Exported to DMN Files

Export decision matrix versions to Decision Model Notation (DMN) files from the Business Rules Engine app. The exported files can be downloaded from the record detail page of a decision matrix version. DMN is a standardized specification for writing business rules. With DMN files, you can reuse matrix versions created in Business Rules Engine in any tool that supports DMN files.

Required Editions

Available in: Lightning Experience
Available in: Enterprise , Unlimited , and Developer Editions for clouds that have Business Rules Engine enabled

User Permissions Needed	
To view and download decision matrix versions exported to DMN files:	Rule Engine Designer

You can view the matrix versions exported to DMN files in the Related tab of a matrix version, only if the version has been exported to a DMN file using connect API.

1. Click , and then find and select **Business Rules Engine**.
2. Click the app navigation menu, and then select **Lookup Tables**.
3. Select a decision matrix.
4. From the Related tab of the matrix, open the version that is to be exported to a DMN file.
5. Click **Export to DMN File**.

When the export is complete, the DMN file is available for download from the Related tab of the version.

6. From the Related tab of the version, select the DMN file.
7. Download the file.

- **Limitations for Exporting Matrix Versions to DMN Files**

Decision matrix versions can be exported to Decision Model Notation (DMN) files from the Business Rules Engine app or by using Connect REST API. Before you attempt to export a matrix version, review these limitations.

Source: https://help.salesforce.com/s/articleView?id=sf.limitations_for_exporting_matrix_versions_to_dmn_files.htm&language=en_US&type=5

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Limitations for Exporting Matrix Versions to DMN Files

Decision matrix versions can be exported to Decision Model Notation (DMN) files from the Business Rules Engine app or by using Connect REST API. Before you attempt to export a matrix version, review these limitations.

Required Editions

Available in: Lightning Experience
Available in: Enterprise , Unlimited , and Developer Editions for clouds that have Business Rules Engine enabled

- Decision matrix versions can't be exported to DMN files if the version columns are of the data type Text Range or Number Range.
- Columns with Currency and Boolean data types are converted to Number data type in the exported file.
- The export fails if the DMN file size exceeds 2 GB.

Source: [https://help.salesforce.com/s/articleView?id=release-
notes.rn_currency_datatype_in_decision_tables.htm&language=en_US&release=244&type=5](https://help.salesforce.com/s/articleView?id=release-notes.rn_currency_datatype_in_decision_tables.htm&language=en_US&release=244&type=5)

Use Currency and Date/Time Data Types in Decision Tables

Create a wider range of rules in decision tables using the Currency and Date/Time data types. With the support for the Currency data type, you can now look up decision tables that use the Currency data type from expression set versions.

Where: This change applies to Lightning Experience in Enterprise, Unlimited, and Developer editions for Industries clouds where Business Rules Engine is enabled.

See Also

[Salesforce Help: Supported Data Types and Operators in Decision Tables](#)

Source: https://help.salesforce.com/s/articleView?id=sf.supported_data_types_and_operators_in_decision_tables.htm&language=en_US&type=5

You are here:

Supported Data Types and Operators in Decision Tables

Decision tables support specific field types as inputs and outputs. A field's data type determines the field's available operators.

Required Editions

Available in: Lightning Experience

Available in: **Enterprise**, **Unlimited**, and **Developer** Editions for clouds that have Business Rules Engine enabled



Note If you use decision tables as part of Loyalty Management or Rebate Management, see [Decision Table](#).

Here are the supported data types and operators in decision tables:

Data Type	Supported Operators
Text	• Equals • Not Equals • Matches • Does Not Match • Contains
Number	• Equals • Not Equals • Less Than • Less or Equal • Greater Than • Greater or Equal
AutoNumber	• Equals • Not Equals • Matches
Percentage	• Equals • Not Equals • Less Than • Less or Equal • Greater Than • Greater or Equal
Date	• Equals • Not Equals • Less Than • Less or Equal

Data Type	Supported Operators
	- Greater Than - Greater or Equal
Date/Time	- Equals - Not Equals - Less Than - Less or Equal - Greater Than - Greater or Equal
Currency	- Equals - Not Equals - Less Than - Less or Equal - Greater Than - Greater or Equal
Checkbox	- Equals - Not Equals
Picklist	- Equals - Not Equals Matches
Picklist (Multi-Select)	- Exists In - Does Not Exist In For multi-select picklist fields, you can pass only one input value during a lookup. For example, imagine you have a multi-select picklist field called "Genre" that stores the genres applicable to a book. This field has the following picklist values: - Genre1 includes ART and LITERATURE - Genre2 includes SCIENCE, FINE ART, and FICTION - Genre3 includes LITERATURE and SCIENCE FICTION When you pass an input value for EXISTS IN: If the input is "Art," there is no match because the lookup is case sensitive. If the input is "ART," it matches Genre1. If the input is "FINE ART," it matches Genre2. If the input is "LITERATURE," it matches both Genre1 and Genre3. When you pass an input value for DOES NOT EXIST IN: If the input is "MYSTERY," it matches all genres because "MYSTERY" is not in any of the picklist values. If the input is "LITERATURE," it matches Genre2 because "LITERATURE" is not in Genre2. If the input is "Art," it matches all genres because the lookup is case sensitive and "Art" does not match "ART."
Lookup	- Equals - Not Equals

Data Type	Supported Operators
Master Detail	• Equals • Not Equals
Email	• Equals • Not Equals • Matches
URL	• Equals • Not Equals • Matches

- **Matches Operator in Decision Tables**

Use the Matches operator to retrieve an output if the query value is an exact match or a subset of the decision table row value. The Matches operator is case-sensitive and applies only to input fields in the decision table.

- **Does Not Match Operator in Decision Tables**

Use the Does Not Match operator to retrieve an output if the query value does not match the decision table row value and is also not a subset of the decision table row value. The Does Not Match operator is case-sensitive and applies only to input fields in the decision table.

- **Contains Operator in Decision Tables**

Use the Contains operator to retrieve an output if the decision table row value is an exact match or subset of the query value. The Contains operator is case-sensitive and applies only to input fields in the decision table.

Source: [https://help.salesforce.com/s/articleView?id=release-
notes.rn_clone_expression_sets.htm&language=en_US&release=244&type=5](https://help.salesforce.com/s/articleView?id=release-notes.rn_clone_expression_sets.htm&language=en_US&release=244&type=5)

Clone Expression Sets

If you want to modify an expression set's versions without affecting the original expression set, clone the expression set. Cloning an expression set clones all the versions in the expression set. You can copy the versions of an expression set that's used as a subexpression in another expression set or is referenced in a flow, without breaking the calling expression set or flow.

Where: This change applies to Lightning Experience in Enterprise, Unlimited, and Developer editions for Industries clouds where Business Rules Engine is enabled.

How: From the record detail page of an expression set, click , and select **Clone**. Enter a new name for the cloned expression set.

See Also

[Salesforce Help: Clone an Expression Set](#)

Loading

✖ Sorry to interrupt

CSS Error

[Refresh](#)

Modal Body...



Change to Device Activation Behavior for SSO [Read More](#)

**We looked high and low
but couldn't find that page.**

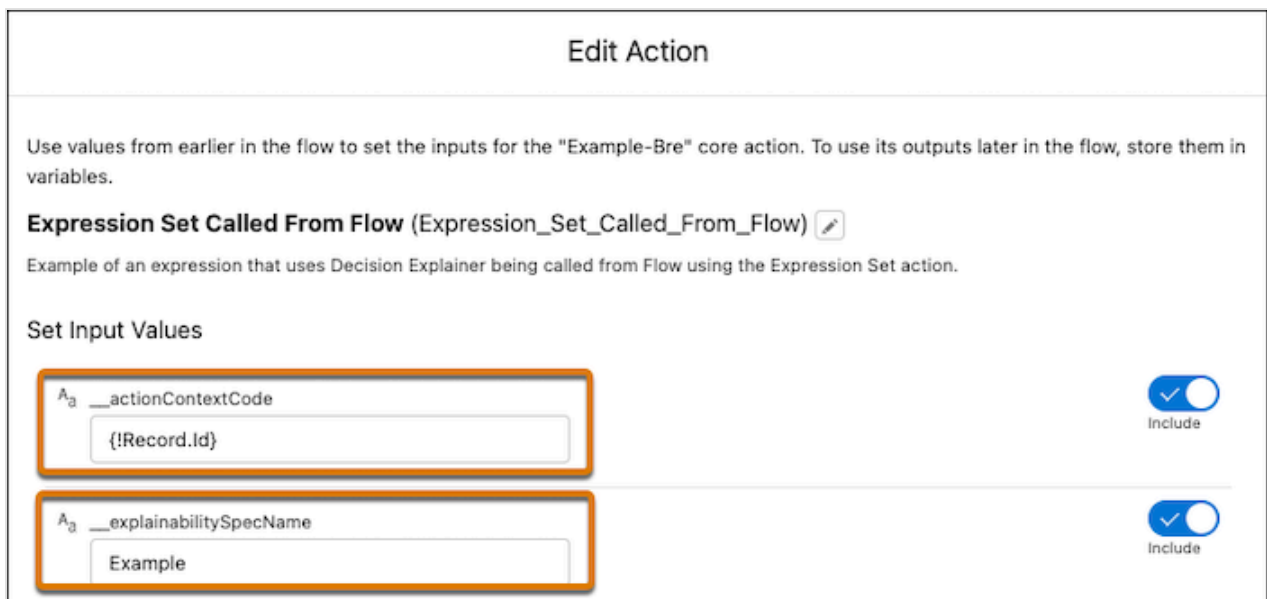
Source: [https://help.salesforce.com/s/articleView?id=release-
notes.rn_action_logs_for_expression_sets_in_flows.htm&language=en_US&release=244&type=5](https://help.salesforce.com/s/articleView?id=release-notes.rn_action_logs_for_expression_sets_in_flows.htm&language=en_US&release=244&type=5)

Capture Action Logs for Expression Sets Called from Salesforce Flow

When a flow calls an expression set that has Decision Explorer enabled, action logs are now generated for every run of the expression set. Action logs show the rationale behind the expression set's output. To capture action logs, you specify the record ID of the object for the `__actionContextCode` parameter, and the explainability action definition or the purpose for which the expression set is used for the `__specName` parameter.

Where: This change applies to Lightning Experience in Enterprise, Unlimited, and Developer editions for Industries clouds where Business Rules Engine is enabled.

How: Select the expression set that you want to call, and specify values for the `__actionContextCode` and `__specName` parameters.



Edit Action

Use values from earlier in the flow to set the inputs for the "Example-Bre" core action. To use its outputs later in the flow, store them in variables.

Expression Set Called From Flow (Expression_Set_Called_From_Flow) 

Example of an expression that uses Decision Explorer being called from Flow using the Expression Set action.

Set Input Values

A₃ <code>__actionContextCode</code>	{!Record.Id}	☒ Include
A₃ <code>__explainabilitySpecName</code>	Example	☒ Include

In this example:

- `{!Record.Id}` represents the flow variable named Record, of the type Record to capture action logs for the expression set in context of an object. The variable Record is related to the Contact object, and `.Id` represents the record ID of the object.
- Example is the name of the Explainability Action Definition against which the expression set is run from Flow.

See Also

[Salesforce Help: Invoke an Expression Set That Uses Decision Explorer from a Flow](#)
[Salesforce Help: Example of Invoking a Decision Explorer Enabled Expression Set from Flow](#)

Source: [https://help.salesforce.com/s/articleView?](https://help.salesforce.com/s/articleView?id=sf.invoke_an_expression_set_that_uses_decision_explainer_from_a_flow.htm&language=en_US&type=5)

[id=sf.invoke_an_expression_set_that_uses_decision_explainer_from_a_flow.htm&language=en_US&type=5](https://help.salesforce.com/s/articleView?id=sf.invoke_an_expression_set_that_uses_decision_explainer_from_a_flow.htm&language=en_US&type=5)

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Invoke an Expression Set That Uses Decision Explorer from a Flow

After an expression set version that has decision explanations enabled is activated, call the expression set that the version belongs to from a flow. To capture action logs for the expression set runs, specify values for the autogenerated `__actionContextCode` and `__specName` parameters in the flow.

Required Editions

Available in: Lightning Experience
Available in: Enterprise , Unlimited , and Developer Editions for clouds that have Business Rules Engine enabled

User Permissions Needed	
To create explainability message templates, explainability action definitions, and explainability action versions:	Decision Explorer Service Access
To enable Decision Explorer for an expression set version:	Rule Engine Designer
To invoke an expression set from Flow	Manage Flows

To enable Decision Explorer for an expression set and add explainability message templates to a version, your Salesforce instance must have Decision Explorer enabled.

1. From Setup, in the Quick Find box, enter `Flows`, and then select **Flows**.
2. Click **New Flow**.
3. Select a flow type, and then click **Create**.
4. Click , and then select **Action**.
5. In the Action search bar, search for, and select the expression set you want to run.
6. Enter a label for the expression set. The API name autopopulates from the label you enter.
7. If needed, enter a description.
8. Set input values for the expression set:
 1. Include `__actionContextCode` and select the variable that captures the record ID of the object for which the expression set is run.
 2. Include `__explainabilitySpecName` and enter the name of the [explainability action definition](#) that defines the purpose for which the expression set is run.



Important Ensure that the explainability action definition you specify has an associated, active [explainability action version](#).

3. If needed, include other variables from the expression set and set their values.

New Action

Filter By
Category

Example-Bre

Use values from earlier in the flow to set the inputs for the "Example-Bre" core action. To use its outputs later in the flow, store them in variables.

* Label: Calling Expression Set Example * API Name: Calling_Expression_Set_Example

Description: Example of how to capture actions logs for every run of an expression set that uses Decision Explorer.

Set Input Values

A_a __actionContextCode	{!ContactRecord.Id}	☒ Include
A_a __explainabilitySpecName	Example	☒ Include
A_a ApplicantId	{!ApplicantBirthdate}	☐ Don't include

Cancel Done

Here, `{!contactRecord.Id}` is the local flow variable to capture the record ID of the contact for whom this flow is run. Example is the explainability action definition, and `{!ApplicantBirthdate}` is the local flow variable that captures the user's input.

9. Save your changes.

10. To verify that the flow works as expected, run or debug the flow using sample values.

EXAMPLE (RUN EXPRESSION SET): Calling Expression Set Example

Inputs:

__actionContextCode = {!ContactRecord.Id}
(003S7000001xTmnIAE)

__explainabilitySpecName = Example

Birthdate = {!ApplicantBirthdate} (March 1, 1955)

Outputs:

DiscountedFees (\$7,000.00)

ApplicantAge (68)

condition_output__3 (true)

condition_output__2 (false)

condition_output__1 (false)

Here, the flow passes the contact record's birthdate to the expression set, which uses the date to calculate the age. The expression set is configured to have \$10,000 as the course fees. Based on the calculated age, the expression set returns the discounted fees of \$7,000 to the flow.

11. Go to the record detail page of the object for which you entered the record ID in the **__actionContextCode** field.



Important To see the decision explainer action logs, the record detail page of the object used as the actionContextCode must have the Decision Explainer Log History component added to the object's record page

Here's the record detail page of the sample contact specified while debugging the flow. The Decision Explainer Log History section captures each run of the expression set.

The screenshot shows the Salesforce Lightning interface for a contact named Ms. Babara Levy. The 'Decision Explorer Log History' section is visible, showing a table with the following data:

Date And Time	Expression Set Name	Version Number
3/20/2023, 04:30 PM	Example V1 CLONE	2
3/20/2023, 04:12 PM	Example V1	1
3/20/2023, 04:14 PM	Example V1	1
3/20/2023, 04:26 PM	Example V2	2
3/20/2023, 04:28 PM	Example V2	2

- To view step-wise details and output of the expression set version called from the flow, select a specific run, and click **View**.

See Also

[Create an Explainability Message Template](#)

[Show Rule Explanations to Users](#)

[Decision Explorer](#)

Source: [https://help.salesforce.com/s/articleView?](https://help.salesforce.com/s/articleView?id=sf.example_invoking_des_enabled_expression_set_from_flow.htm&language=en_US&type=5)

[id=sf.example_invoking_des_enabled_expression_set_from_flow.htm&language=en_US&type=5](https://help.salesforce.com/s/articleView?id=sf.example_invoking_des_enabled_expression_set_from_flow.htm&language=en_US&type=5)

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Example of Invoking a Decision Explainer Enabled Expression Set from Flow

This example shows how to invoke an expression set that uses Decision Explainer from flow. The example also shows how to view every run of the expression set in the action logs for the context in which the expression set is run.

Required Editions

Available in: Lightning Experience

Available in: **Enterprise**, **Unlimited**, and **Developer** Editions for Industries clouds that have Business Rules Engine and Decision Explainer enabled.

To promote fitness and well-being, a company is offering a discount on its annual gym membership to its contacts. To encourage physical movement, the company launched a week-long step challenge for its contacts. The annual membership price is \$1,000. A discount is offered on this price based on a contact's age and the step count from the week-long step challenge.

The company's sales team gave the rule designer and the Salesforce admin the age and step count ranges and the applicable discount percentages. The company also asked that the discounted gym membership fees be updated on a contact's record detail page for quick reference.

The rule designer contacted the company's Salesforce admin and together they divided the requirement into these tasks:

Task	Description
Create a decision matrix to calculate the discount percent based on the step count from the step challenge.	The matrix uses an input column with the datatype number range to capture the step count. The matrix captures the discount percent in the output column for these ranges.
Create a decision matrix to get age-based discount.	The matrix uses an input column with the data type number range to define the age range. The matrix captures the discount percent in the output column for these ranges.
Create an expression set and its version to design the end-to-end rule to calculate the discounted gym membership.	The expression set version: • Takes the step count as an input and looks up the decision matrix to get the step-based discount. • Uses the contact's birth date to calculate age. • Passes the contact's age to the age-based discount matrix to get the additional discount.

Task	Description
	- Calculates the total discount by adding the age-based and the step-based discounts. - Calculates the discounted premium by subtracting the total discount percent from the base membership fee of \$1,000. - Enables Decision Explainer for customer service representatives to understand the logic behind the discounted membership fee for a specific contact.
Create a flow to run the expression set and return the output to the flow.	- The flow captures the contact's step count and passes it to the expression set action. - The expression set action is configured to use a contact's record ID as the context in which the flow and the expression set run. - The expression set action returns the discounted membership fees to the flow. - The flow then updates the contact record with the discounted fees.
To launch the flow for a contact, add the flow action to the contact record detail page.	- The company's customer service representatives can launch the flow for a contact, and enter the contact's step count for the previous week. - The flow runs in the background and updates the Gym Membership Fees field for the contact record.
View the action log for the flow.	The customer service associates can see the Decision Explainer Log History on the contact's record detail page to check how the gym membership fees for the contact were calculated.

- **Example: Create a Decision Matrix to Get Step-Based Discount Percent**

Create a decision matrix and configure its version with an input column for number of steps, and an output column with the corresponding discount percent based on a number range of steps.

- **Example: Create a Decision Matrix to Get Age-Based Discount**

Create a decision matrix and configure its version with an input column for age, and an output column with corresponding discount percent based on the age range.

- **Example: Create an Expression Set to Calculate Discounted Membership Fees**

Configure the expression set version to calculate a contact's age from their birth date, look up the contact's step count to get the steps-based discount, and return the discounted gym membership fee after subtracting the total discount from the base membership fee.

- **Example: Create a Flow to Run the Expression Set and Update the Record**

Create an input form to capture a contact's step count and pass it to the expression set action. Configure the flow to update the contact record with the expression set's output.

- **Example: Add the Flow to the Contact Record Page and Run the Flow**

To enter the step count for a contact from the contact's record page, and pass the contact's record ID to the flow from the same page, add a flow component to the Contact Record Page.

- **Example: View Decision Explainer Log History**

View the action log for the expression set that was run from the flow for a contact record.

See Also

[Create an Explainability Message Template](#)

[Show Rule Explanations to Users](#)

[Show Rule Explanations to Users \(Example\)](#)

[Decision Explainer for Expression Sets](#)

Source: https://help.salesforce.com/s/articleView?id=sf.decision_explainer.htm&language=en_US&type=5

You are here:

Decision Explorer for Expression Sets

Use Decision Explorer with Business Rules Engine to show your users explanations for the results of steps in an expression set, and to capture action logs for expression set runs. Store explanations for expression steps in explainability message templates. And then add the templates to your expression set steps for which you want to show explanations. When you simulate the expression set version, switch to the Concise view option to preview how your users see the explanations.

Required Editions

Available in: Lightning Experience
Available in: Enterprise , Unlimited , and Developer Editions for Industries clouds that have Business Rules Engine and Decision Explorer enabled

Decision Explorer is a service that provides insight into why a business rule generated a specific result. For example, a student wants to know why they received a lesser discount on their course fee than what they expected. With Decision Explorer, the student application reviewer can check the application's history and the expression set that's used to evaluate the application, and provide the student the reason for receiving a lesser discount.

- **Create Expression Set Message Tokens**

Expression set message tokens serve as placeholders for specific values from an expression set version within an explainability message template. Without tokens, the explanation messages would be static and unable to reflect the specific data involved in a particular execution of the rule. Create a token, and then map it to actual resources to dynamically display relevant data in explanation messages.

- **Create an Explainability Message Template**

Explainability message templates store explanation messages, and can be referenced in an expression set version step to explain the step's result. You can use expression set message tokens in your message templates as placeholders for expression set version resources. Rule designers can then map the tokens to the resources from the expression set version where the template is used. When the expression set runs, the explanations dynamically show the values of the mapped resources.

- **Show Rule Explanations to Users**

When Decision Explorer is available in your Salesforce org, you can show your users the explanations for expression set version step results. The explanations are stored in explainability message templates that your admin creates and maintains.

- **Disable Decision Explanations for Expression Set Version Steps**

To remove the explainability message templates assigned to an expression set version step, open the version in Expression Set Builder and deselect Show decision explanation

for the step. You can disable decision explanations for expression set version steps only if the version is inactive.

- **Show Rule Explanations to Users (Example)**

This example shows how you can configure an expression set version to show step-result explanations to external users. The example also shows how to preview the explanations that your users see.

- **Compare Expression Set Runs with Decision Explainer**

To analyze why an expression set generated different outputs for each run, enable Decision Explainer for expression set versions, and then use the Compare option on the Decision Explainer Log History component. Decision Explainer logs are generated in the context of object records that an expression set is run for. To use the Compare option, the Decision Explainer Log History component must be enabled on the record page of the object within whose context the expression set is run.

- **Explainability Message Templates for Developers**

Developers in your Salesforce org can use developer resources to use and extend explainability message templates.

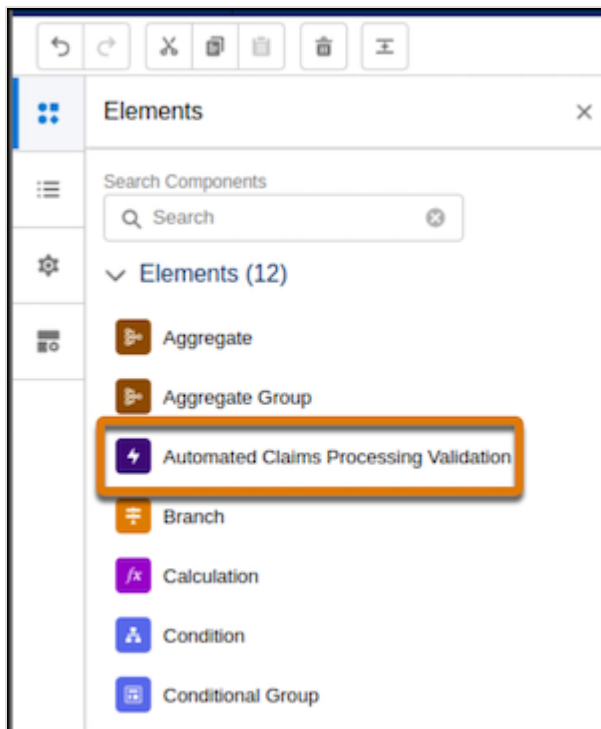
Source: [https://help.salesforce.com/s/articleView?id=release-
notes.rn_business_elements_as_first_level_elements_in_builder.htm&language=en_US&release=244&type=5](https://help.salesforce.com/s/articleView?id=release-notes.rn_business_elements_as_first_level_elements_in_builder.htm&language=en_US&release=244&type=5)

View Business Elements as First-Level Elements in Expression Set Builder

Business elements shipped with your cloud are now listed as first-level elements and look like standard step elements. Previously, you had to add a business element as a step on the builder canvas, and then search for and select the business element from the available list.

Where: This change applies to Lightning Experience in Enterprise, Unlimited, and Developer editions for Industries clouds where Business Rules Engine is enabled.

How: Open an expression set version in the builder, and click + to view and add business elements. You can also view the elements by clicking  on the builder canvas.



Source: [https://help.salesforce.com/s/articleView?id=release-
notes.rn_simulate_expression_set_versions_with_json_input.htm&language=en_US&release=244&type=5](https://help.salesforce.com/s/articleView?id=release-notes.rn_simulate_expression_set_versions_with_json_input.htm&language=en_US&release=244&type=5)

Simulate Expression Set Versions Using JSON Input

Want to simulate an expression set version that has a large number of input variables? You can now pass the input data in JSON format using the Advanced input mode for simulation. With the Advanced input mode, you can enter the input values in the JSON input text area. Previously, all the input variables in the version only appeared as fields, and you entered values for each field. The JSON input option simplifies and speeds up the process of passing input variables, especially when you have a large number of values to input.

Where: This change applies to Lightning Experience in Enterprise, Unlimited, and Developer editions for Industries clouds where Business Rules Engine is enabled.

How: On the simulation screen, for Input Mode, select **Advanced**. Replace placeholders in the text area with the values you want to use for the simulation. When you simulate the expression set version again, the values you enter are retained in the text area. To simulate using different values, modify the JSON input.

Simulation Details

Input

Output

Input Variables

Enter input variable values to simulate your expression set.

Effective Date Range

From Nov 23, 2020 to Nov 14, 2024

Simulation Date

Date

Mar 27, 2023

Time

3:58 AM

* Input Mode

Advanced

Download Sample JSON Input

JSON Input

```
{
  "productColor": "<Text>",
  "productName": "<Text>"
}
```

See Also

[Salesforce Help: Simulate and Activate Your Expression Set Version](#)

Source: https://help.salesforce.com/s/articleView?id=sf.simulate_and_activate_the_expression_set.htm&language=en_US&type=5

You are here:

Simulate and Activate Your Expression Set Version

Before you activate your expression set, run simulations with default and custom test input variables. Different decision matrices or subexpression versions called in an expression set version can have different start date time and end date time values. Test different versions by changing the Effective Date value. If the expression set doesn't work as expected, edit the elements and resimulate. When you're satisfied, activate the expression set version. If the expression set version is related to a context definition, you can simulate it by using the context mappings available in the context definition.

Required Editions

Available in: Lightning Experience
Available in: Enterprise , Unlimited , and Developer Editions for clouds that have Business Rules Engine enabled

User Permissions Needed	
To create, update, and delete expression sets:	Rule Engine Designer
To use expression sets in Business Rules Engine:	Rule Engine Runtime

Simulate the StudentDiscountCalculation expression set that we created to test if the expression set is working as intended, and then activate the set.

1. Open your expression set version in Expression Set Builder.
2. Click **Simulate**.
The Input Mode dropdown displays the options for providing sample simulation data to the version.
3. To enter sample values for the expression set version variables in the Variables section, select **Simplified** Input Mode.

Input

Output

Input Variables
Enter input variable values to simulate your expression set.

Effective Date Range
Mar 23, 2023 onwards

Simulation Date
Date
Apr 4, 2023
Time
10:32 AM

Input Mode
Simplified

Variables

GPA
Number

QuoteAmount
Currency

StudentBirthdate
Date

Reset

Simulate

4. To enter sample values for the expression set variables in the JSON Input text box, select **Advanced** Input Mode.

Simulation Details

Input
Output

Input Variables

Enter input variable values to simulate your expression set.

Effective Date Range

Jul 18, 2024, 10:06 PM onwards

Simulation Date

Date
Time

Aug 16, 2024
2:38 AM

MMM d, yyyy

* Input Mode

Advanced

Context Definition

Salestransaction_clone

* Context Mapping

QuoteEntitiesMapping

[Download Sample JSON Input](#)

JSON Input

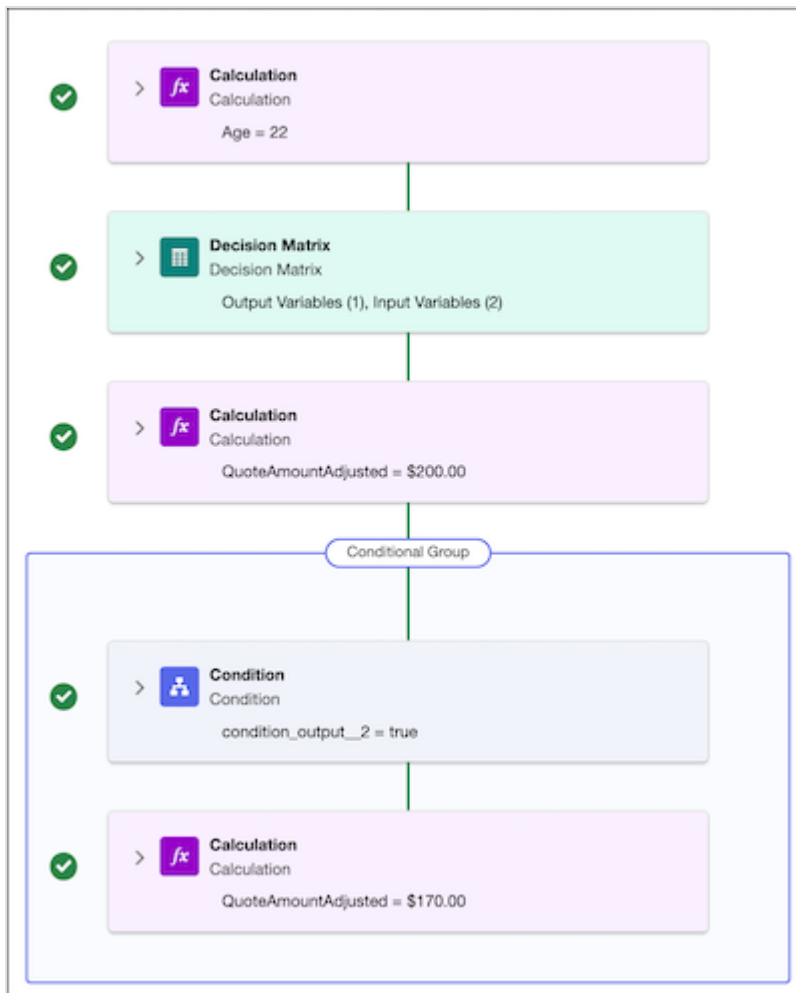
```

{
  "AssetTag": [
    {
      "AssetActionTag": [
        {
          "TotalQuantity": "10",
          "AssetActionAmount": "20"
        }
      ],
      "CurrentQuantity": "10"
    }
  ]
}

```

Reset
Simulate

1. Change the selection to a mapping of your choice.
The Context Mappings dropdown lists the available options if the expression set is linked to a context definition. The default context mapping is selected.
2. Enter sample values for the variables in the JSON Input text box.
Alternatively, download the sample JSON input file, modify the file with your input values, and paste the updated JSON inputs in the textbox.
5. Simulate the expression set.
Here's how the simulation results look in Detailed View, which is the default simulation view.



Tip To see how the results appear to your users, switch to Concise View. Concise View shows the explanations for version steps where you added explainability message templates.

6. For comprehensive testing of the expression set, enter different variable values, change the mapping, and resimulate.
7. If you're happy with the simulation result, activate the expression set version.

See Also

[Salesforce Help: Considerations for Calling Expression Sets](#)

[Salesforce Help: Invoke an Expression Set in a Flow](#)

[Salesforce Help: Call an Expression Set from an Integration Procedure](#)

[Salesforce Help: Call an Expression Set from an OmniScript](#)

Source: [https://help.salesforce.com/s/articleView?id=release-
notes.rn_changed_objects_in_business_rules_engine.htm&language=en_US&release=244&type=5](https://help.salesforce.com/s/articleView?id=release-notes.rn_changed_objects_in_business_rules_engine.htm&language=en_US&release=244&type=5)

Changed Object in Business Rules Engine

Do more with this changed Business Rules Engine object.

View the status of exporting a decision matrix version to a DMN file

Use the new `DscnModelNoteExportStatus` field in the `CalculationMatrixVersion` object.

Source: [https://help.salesforce.com/s/articleView?id=release-
notes.rn_business_rules_engine_connect_rest_apis.htm&language=en_US&release=244&type=5](https://help.salesforce.com/s/articleView?id=release-notes.rn_business_rules_engine_connect_rest_apis.htm&language=en_US&release=244&type=5)

Connect REST APIs

Connect REST APIs for Business Rules Engine help customers, partners, and ISVs integrate with Salesforce software and UIs.

Create an expression set in Business Rules Engine

Make a POST request to the new `/connect/business-rules/expression-set` resource.

New request body: Expression Set Input

New response body: Expression Set

Export a decision matrix to a Decision Model Notation (DMN) file

Make a POST request to the new `/connect/business-rules/decision-models/export` resource.

New request body: Decision Model Export Input

New response body: Decision Model Export Output

Perform a lookup on a decision table

Make a POST request to the new `/connect/business-rules/decision-table/lookup/` resource.

New request body: Decision Table Input

New response body: Decision Table Outcome

Invoke a decision table by passing multiple input conditions within the same request

Make a POST request to the new `/connect/business-rules/decision-table/lookup/${decisionTableId}` resource.

New request body: Decision Table Bulk Input

New response body: Decision Table Bulk Outcome

Create a decision table definition in Business Rules Engine

Make a POST request to the new `/connect/business-rules/decision-table/definitions` resource.

New request body: Decision Table Definition Input

New response body: Decision Table Output

Get and delete a decision table definition

Make a GET and DELETE request to the new `/connect/business-rules/decision-table/definitions/${decisionTableId}` resource.

New response body: Decision Table Output

Update a decision table definition

Make a PATCH request to the new `/connect/business-rules/decision-table/definitions/${decisionTableId}` resource.

New request body: Decision Table Definition Input

New response body: Decision Table Output

Retrieve dependencies of an expression set in Business Rules Engine

Make a GET request to the new

```
/connect/business-rules/expression-set/version/${expressionSetVersionId}/dependencies
```

resource.

New response bodies: Dependency Detail Output and Expression Set Version Dependency.

See Also

[Salesforce Developer Guide: Business Rules Engine Business APIs](#)

Source: [https://help.salesforce.com/s/articleView?id=release-
notes.rn_bre_new_metadata_api.htm&language=en_US&release=244&type=5](https://help.salesforce.com/s/articleView?id=release-notes.rn_bre_new_metadata_api.htm&language=en_US&release=244&type=5)

New and Changed Business Rules Engine Metadata Types

Make the most of the new and changed metadata types of Business Rules Engine.

Metadata Types

Determine whether the decision explanation is exposed to external users

Use the new `shouldShowExplExternally` field on the `ExpressionSetDefinitionVersion` subtype of the `ExpressionSetDefiniton` metadata type. Introduced in API version 56.0, this type has been added to the *Industries Common Resources Developer Guide*.

Incorporate custom elements into an Expression Set.

Use the new `ExpressionSetCustomElement` subtype of the `ExpressionSetStep` metadata type on the existing `ExpressionSetDefinition` metadata type. Introduced in API version 56.0, this type has been added to the *Industries Common Resources Developer Guide*.

Specify a new result type in the message template

Use the new `NoResult` value in the existing `evaluationResult` field on the `ExplainabilityMsgTemplate` metadata type.

Specify new step types in an expression set in explainability message template.

Use the new `ListEnabledGroup`, `ListFilter`, and `ReferenceProcedure` values in the existing `expressionSetStepType` field on the `ExplainabilityMsgTemplate` metadata type.